

Statistical Modeling

Part II

Recorded from the lecturer in 2026-2027.

What is a statistical model?

Data modelling vs. Algorithmic modelling
Inference vs. Computing

Def. A statistical model is

$$\{P_\theta : \theta \in \Theta\}.$$

Examples.

1.

$$\mathbb{E}\{Y \mid X = x\} = x^\top \beta.$$

2.

$$Y = X^\top \beta + \varepsilon, \quad \varepsilon \mid X \sim N(0, \sigma^2).$$

3.

$$Y(x) = x^\top \beta + \varepsilon.$$

Probability

$$(\Omega, \mathcal{F}, P).$$

Random variable/vector.

- CDF:

$$F(x) = P(X \leq x), \quad x \in \mathbb{R}.$$

- PDF/PMF:

$$f(x) = \frac{dF(x)}{dx} \quad \text{or} \quad f(x) = P(X = x).$$

- Quantile function:

$$F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}, \quad u \in [0, 1].$$

- MGF:

$$M(t) = \mathbb{E}\{e^{tX}\}, \quad \text{for } t \text{ in a neighbourhood of } 0.$$

Independence. Expectation. Variance.

Random

$$X = (X_1, \dots, X_n) \in \mathbb{R}^n.$$

Fixed

$$A \in \mathbb{R}^{m \times n}, \quad b \in \mathbb{R}^m.$$
$$\mathbb{E}\{AX + b\} = A\mathbb{E}\{X\} + b.$$

Normal distribution

$$Z \sim N_d(\mu, \Sigma), \quad \mu \in \mathbb{R}^d, \quad \Sigma \in \mathbb{R}^{d \times d} \text{ positive definite.}$$

$$f(z) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} e^{-\frac{1}{2}(z-\mu)^\top \Sigma^{-1}(z-\mu)}, \quad z \in \mathbb{R}^d.$$

Properties.

1.

$$AZ + b \sim N(A\mu + b, A\Sigma A^\top).$$

2. If

$$\begin{pmatrix} Z_1 \\ Z_2 \end{pmatrix}$$

is normal, then

$$Z_1 \perp\!\!\!\perp Z_2 \iff \text{Cov}(Z_1, Z_2) = 0.$$

Informally:

$$\chi_d^2 = \underbrace{N(0, 1)^2 + \dots + N(0, 1)^2}_{d \text{ times}}.$$

$$t_d = \frac{N(0, 1)}{\sqrt{\chi_d^2/d}}, \quad F_{d_1, d_2} = \frac{\chi_{d_1}^2/d_1}{\chi_{d_2}^2/d_2}.$$

Def.

- We say $\mathcal{P}$ is *identifiable* if

$$P_{\theta_1} = P_{\theta_2} \quad \text{iff} \quad \theta_1 = \theta_2.$$

- We say $\mathcal{P}$ is *parametric* if Θ is finite-dimensional ($\Theta \subseteq \mathbb{R}^p$). Otherwise non-/semi-parametric.

Conditional expectation

$$\mathbb{E}\{Y \mid X = x\}.$$

Law of total prob./exp.

$$\mathbb{E}\{Y\} = \mathbb{E}[\mathbb{E}\{Y \mid X\}].$$

Estimation

Likelihood function

$$L(\theta) = f(\underline{X}; \theta).$$

i.i.d.

Def. MLE:

$$\hat{\theta} = \arg \max_{\theta \in \Theta} L(\theta) \quad \text{is a statistic.}$$

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$$\text{Bias}_\theta(\hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta.$$

•

$$\text{MSE}_\theta(\hat{\theta}) = \mathbb{E}_\theta\{(\hat{\theta} - \theta)^2\} = \text{Bias}_\theta(\hat{\theta})^2 + \text{Var}_\theta(\hat{\theta}).$$

- $T(X)$ is *sufficient* if $X \mid T$ is independent of θ .

- **Rao–Blackwell**

$$\text{MSE}_\theta \left(\hat{\theta} = \mathbb{E}(\tilde{\theta} \mid T) \right) \leq \text{MSE}_\theta(\tilde{\theta}), \quad \text{for all } \theta.$$

- $S(X) \subseteq \Theta$ is a $(1 - \alpha)$ -confidence set if

$$P_\theta\{\theta \in S(X)\} = 1 - \alpha, \quad \forall \theta \in \Theta.$$

Typically via pivotal quantity $R(X, \theta)$, distribution independent of θ .

Hypothesis testing

$$H_0 : \theta \in \Theta_0 \quad \text{vs.} \quad H_1 : \theta \in \Theta_1 \quad (\Theta_0, \Theta_1 \text{ disjoint}).$$

Test function

$$T(X) \in \{0, 1\}.$$

Neyman–Pearson theory

Def.

- Power function

$$\beta(\theta) = P_\theta(T(X) = 1).$$

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$$\text{Size} = \sup_{\theta \in \Theta_0} \beta(\theta).$$

$$\begin{aligned} \max. \quad & \beta(\theta) \quad \text{over } \Theta_1, \\ \text{s.t.} \quad & \text{size} \leq \alpha. \end{aligned}$$

N–P Lemma

If

$$\Theta_0 = \{\theta_0\}, \quad \Theta_1 = \{\theta_1\},$$

solution is

$$T(X) = \mathbf{1} \left\{ \left\lfloor \frac{L(\theta_1)}{L(\theta_0)} \right\rfloor \geq c(\alpha) \right\}.$$

Generalized LR:

$$\Lambda(X) = \frac{\sup_{\theta \in \Theta_0 \cup \Theta_1} L(\theta)}{\sup_{\theta \in \Theta_0} L(\theta)} \quad \text{LR statistic.}$$

$$\begin{array}{ll} \text{multinomial} & / \text{P. } \chi^2 \\ \text{normal} & / \chi^2, F. \end{array}$$

Duality

If

$$T(\theta, X) = \mathbf{1}\{\theta \notin S(X)\}, \quad S(X) = \{\theta : T(\theta, X) = 0\},$$

then $T(\theta_0, X)$ is size- α of $H_0 : \theta = \theta_0$ for all θ_0 iff $S(X)$ is a $(1 - \alpha)$ -confidence set.

end of lecture 1